

A column-square estimate for four-generated numerical semigroups

Research draft v0.2 — 23 September 2026

For independent mathematical review. Novelty has not been established. Formal verification covers the new combinatorial argument; see Section 5.

Abstract

We use a single terminal Hilbert–Samuel budget to control the sum of the squares of the column lengths of a two-variable lex section. Completing a square before applying Cauchy–Schwarz preserves the position of the last nonzero column degree and yields a strict quadratic bound for every integer width $w \geq 4$. This replaces the finite enumeration in draft v0.1 and removes the use of a separate large-width estimate. Combined with published algebraic reductions, the argument gives the width bounds for four-generated numerical semigroup rings, with a strict first bound when $w \geq 4$. The new combinatorial proof, including the uniform signed-vector inequality, is verified in Lean. The algebraic transfer and the novelty assessment remain subject to independent review.

1 The semigroup statement

A numerical semigroup is an additive submonoid $\Gamma \subseteq \mathbb{N}$ with finite complement, where $\mathbb{N} = \{0, 1, \dots\}$. Suppose Γ has exactly four minimal generators $g_0 < g_1 < g_2 < g_3$, and put $m = g_0$ and $w = g_3 - g_0$. For a field $\mathbb{k}$, write

$$R_\Gamma = \mathbb{k}[[t^{g_0}, t^{g_1}, t^{g_2}, t^{g_3}]].$$

All Betti numbers of R_Γ are over its minimal regular presentation $\mathbb{k}[[X_0, X_1, X_2, X_3]]$.

Theorem 1.1 *For every such Γ and every field $\mathbb{k}$,*

$$\beta_i(R_\Gamma) \leq i \binom{w+1}{i+1} \quad (i \geq 1).$$

If $w \geq 4$, then in addition

$$\beta_1(R_\Gamma) \leq \binom{w+1}{2} - 1.$$

Caviglia–Moscariello–Sammartano (CMS) prove the non-strict bounds for $w \geq 40$ in [1, Theorem 1.5]; their Remark 5.2 discusses the remaining small-width lex problem. Draft v0.1 treated that interval using a finite envelope certificate. The present proof works uniformly for all $w \geq 4$ and makes the first bound strict. The case $w = 3$ remains the known consecutive-generator case. We do not claim these stronger conclusions for arbitrary tangent cones or for arbitrary embedding dimension.

2 A terminal budget for lex columns

Let $S = \mathbb{k}[x, y, z]$, ordered lexicographically with $x > y > z$, and let $L \subseteq (x, y, z)^2$ be an artinian lex ideal. Its standard monomials are those outside L . Assume that, for an integer $w \geq 4$,

$$\sum_{t=0}^d \text{HF}(S/L, t) \leq 1 + dw \quad (d \in \mathbb{N}). \quad (1)$$

Put $K = L \cap \mathbb{k}[x, y]$, and let α be the least exponent such that $x^\alpha \in L$. Then $\alpha \geq 2$. For $0 \leq i < \alpha$, define the positive column length

$$n_i = \#\{j \geq 0 : x^i y^j \notin K\}, \quad \ell = \text{length}(\mathbb{k}[x, y]/K) = \sum_{i=0}^{\alpha-1} n_i.$$

Each column is an initial interval in its y exponent. Since $x^i y^{n_i-1}$ is standard, lex order makes y^{i+n_i-1} standard. Hence

$$i + n_i \leq n_0 \quad (0 \leq i < \alpha). \quad (2)$$

Lemma 2.1 *The column lengths satisfy*

$$\sum_{i=0}^{\alpha-1} \frac{n_i(n_i + 1)}{2} \leq 1 + (n_0 - 1)w. \quad (3)$$

Proof. For fixed i , all monomials $x^i y^j z^k$ with $j + k < n_i$ are standard: first $x^i y^{j+k}$ is standard, and replacing some y factors by z lowers lex order. They have total degree at most $i + n_i - 1 \leq n_0 - 1$, by (2). Their number is $n_i(n_i + 1)/2$, and the sets for distinct i are disjoint. Apply (1) once, at $d = n_0 - 1$. $\square$

Lemma 2.2 *One has $\alpha \leq w - 2$.*

Proof. All monomials of degree below α are standard, since the largest such monomials x^d are standard. Thus

$$\binom{\alpha + 2}{3} \leq 1 + (\alpha - 1)w.$$

Using

$$\alpha(\alpha + 1)(\alpha + 2) - 6 = (\alpha - 1)(\alpha^2 + 4\alpha + 6)$$

and $\alpha \geq 2$ gives $\alpha^2 + 4\alpha + 6 \leq 6w$. If $\alpha \geq w - 1$, this implies $w^2 + 2w + 3 \leq 6w$, or $(w - 1)(w - 3) \leq 0$, impossible for $w \geq 4$. The integer parameters therefore satisfy $\alpha \leq w - 2$. $\square$

3 A uniform strict estimate

Lemma 3.1 *Under (3),*

$$(2\ell + \alpha - 2w)^2 \leq \alpha[4(w - 1)(w - 2) + \alpha]. \quad (4)$$

Proof. Use the signed integer vector

$$v_0 = 2n_0 + 1 - 2w, \quad v_i = 2n_i + 1 \quad (1 \leq i < \alpha).$$

There is no nonnegativity assumption on v_0 . Direct expansion and (3) yield

$$\begin{aligned}\sum_i v_i &= 2\ell + \alpha - 2w, \\ \sum_i v_i^2 &= 4 \sum_i n_i(n_i + 1) + \alpha - 8wn_0 - 4w + 4w^2 \\ &\leq 4(w-1)(w-2) + \alpha.\end{aligned}$$

Now apply $(\sum_i v_i)^2 \leq \alpha \sum_i v_i^2$. Equivalently, this step follows from the integer identity

$$\alpha \sum_i v_i^2 - (\sum_i v_i)^2 = \sum_{i < j} (v_i - v_j)^2 \geq 0.$$

□

Theorem 3.2 *For every integer $w \geq 4$ under the hypotheses of Section 2,*

$$\alpha + 1 + \ell \leq \binom{w+1}{2} - 1. \quad (5)$$

Moreover,

$$\alpha + 2\ell \leq 2\binom{w+1}{3}, \quad \ell \leq 3\binom{w+1}{4}. \quad (6)$$

Proof. Set $T = w^2 - w - \alpha - 2$. Lemma 2.2 gives $T \geq w(w-2) > 0$. A polynomial identity and the same lemma show

$$\begin{aligned}T^2 - \alpha[4(w-1)(w-2) + \alpha] &= (w-2)[(w-2)(w+1)^2 - 2\alpha(3w-1)] \\ &\geq (w-2)^2(w-1)(w-3) > 0.\end{aligned}$$

Combining this with (4) gives $2\ell + \alpha - 2w < T$, since $T > 0$. Rearranging yields

$$\alpha + 1 + \ell < \frac{w(w+1)}{2},$$

which is (5) by integrality.

Write $C = \binom{w+1}{2}$. Then $\ell \leq C - \alpha - 2$, so $\alpha + 2\ell \leq 2C - \alpha - 4 \leq 2C$ and $\ell \leq C$. Finally,

$$C \leq \binom{w+1}{3}, \quad C \leq 3\binom{w+1}{4} \quad (w \geq 4),$$

as is immediate from the ratios $(w-1)/3$ and $(w-1)(w-2)/4$. This proves (6) without a bounded parameter search. □

Remark 3.3 *The retained estimate (4) also gives*

$$\ell \leq w - \frac{\alpha}{2} + \sqrt{\alpha(w-1)(w-2) + \frac{\alpha^2}{4}}.$$

Since $\alpha^2 + 4\alpha + 6 \leq 6w$, this is $O(w^{5/4})$ as w grows. The square-root formulation and asymptotic notation are ordinary consequences recorded here; the Lean conclusion used below is the exact integer bound (5)–(6).

4 The algebraic transfer

For the finite-colength lex ideal L , the two-variable lex ideal K has $\alpha + 1$ minimal generators and rank one, so its Betti numbers as an ideal are $(\alpha + 1, \alpha)$. The finite-colength formula [1, equation (7)] gives

$$\beta_0^S(L) = \alpha + 1 + \ell, \quad \beta_1^S(L) = \alpha + 2\ell, \quad \beta_2^S(L) = \ell. \quad (7)$$

Thus Theorem 3.2 bounds the three positive Betti numbers of S/L , with the usual ideal-to-quotient shift by one.

Proof of Theorem 1.1. The four-generator hypothesis gives $w \geq 3$. If $w = 3$, the generators are consecutive. Proposition 2.7 of [2], with $r = 4$, gives the desired bound for the graded semigroup ring. Localizing its minimal weighted-graded free resolution at the homogeneous maximal ideal and then completing preserve exactness and minimality, hence preserve the Betti numbers.

Suppose now $w \geq 4$. CMS’s construction starts with the Artinian reduction $R_\Gamma/(t^m)$ and passes to its associated graded ring and then to a homogeneous monomial initial ideal $J \subseteq Q = \mathbb{k}[x, y, z]$. It has colength m and no linear forms. The regular parameter t^m , the initial-ideal comparison and [1, equation (1)] give

$$\beta_i(R_\Gamma) \leq \beta_i^Q(Q/J) \quad (i \geq 1).$$

If $w \leq m - 2$, Theorem 3.3 of [1] gives $\text{HS}(Q/J, d) \leq 1 + dw$. If $w \geq m - 1$, the same inequality follows directly: the degree-zero contribution is one, and for $d \geq 1$,

$$\text{HS}(Q/J, d) \leq m \leq w + 1 \leq 1 + dw.$$

Thus the budget holds for every $w \geq 4$, without invoking a separate large-width theorem.

Let $L = \text{Lex}(J)$. It remains artinian, has no linear forms and has the same Hilbert function. The characteristic-independent Bigatti–Hulett–Pardue comparison [1, Theorem 2.1] implies

$$\beta_i(R_\Gamma) \leq \beta_i^Q(Q/J) \leq \beta_i^Q(Q/L).$$

Apply (7) and Theorem 3.2 for $i = 1, 2, 3$. For $i \geq 4$, the Betti numbers vanish: R_Γ is one-dimensional Cohen–Macaulay with a four-dimensional minimal regular presentation, and therefore has projective dimension three. This proves both conclusions. $\square$

The Artinian assumption in (7) is not inferred from the combinatorial theorem. It comes from the semigroup reduction. For example, (x^2, xy, xz, y^2, yz) retains infinitely many standard powers of z and does not satisfy the finite-colength Betti formula. The general lex statement at $w = 3$ is also false; that boundary is handled separately for semigroups.

5 Formal verification, evidence and limits

The new Lean entry points are

```
WidthBounds.all_widths_analytic_bounds
WidthBounds.all_widths_strict_improvement.
```

They work with a predicate on exponent triples, divisor closure, downward same-degree lex closure, the initial pure- x degree, and the cumulative three-variable counting budget. They have no assumption $w \leq 39$ and no assumed column-square or Cauchy conclusion. Column

support, exact total length, the terminal triangle injection, prefix counting, signed-vector arithmetic and strict comparison are all proved.

The proof uses Lean 4.22.0 and mathlib v4.22.0. Its axiom dependencies are the standard `propext`, `Classical.choice` and `Quot.sound`; there are no proof placeholders or new axioms. A separate diagnostic traverses actual declaration dependencies and rejects a dependency on the old finite envelope certificate or small-width theorems. Those older declarations remain available in the project but are not used by the new proof.

The polynomial-ideal interpretation has a separately verified membership interface. The identifications of monomial counts with quotient Hilbert functions and colength, the Betti formula (7), and the full semigroup transfer are conventional algebraic arguments, not an end-to-end Lean formalization. The direct colength argument for $w \geq m - 1$ in Section 4 is part of that conventional layer.

Run `lake build` inside the project's `lean/` directory. The modules `ColumnBounds`, `AnalyticArithmetic`, `AllWidths` and `DependencyAudit` contain the new development. The dependency manifest, exact source hashes and build log accompany the project. Small numerical experiments served to look for mistakes; no such experiment is used in the proof.

Research status. This is a user-directed AI-assisted research draft. The analytic argument received an internal proof review, and its combinatorial formalization has passed Lean's kernel checks. A bounded primary-source search did not locate the same full-range strict conclusion, but that is not a novelty certification. Human domain-expert review and a fuller novelty assessment remain outstanding. Draft v0.1 and its finite certificate are preserved as an independent earlier route.

References

- [1] G. Caviglia, A. Moscariello and A. Sammartano, *Bounds for syzygies of monomial curves*, Proc. Amer. Math. Soc. **152** (2024), 3665–3678. doi:10.1090/proc/16862. Version used: arXiv:2307.05770v2, 20 July 2024.
- [2] J. Herzog and D. I. Stamate, *On the defining equations of the tangent cone of a numerical semigroup ring*, J. Algebra **418** (2014), 8–28. doi:10.1016/j.jalgebra.2014.07.008. Version used: arXiv:1308.4644v3, 29 July 2014.